

Practical Assignment: 4

Practical Task Instructions

- **Execute the given code** and take a **screenshot of the error** displayed during execution.
- **Identify and explain the error**, including the reason why it occurs.
- **Justify your explanation** by referring to the relevant Java concept or rule.
- **Correct the code** and execute it again to verify that it works successfully.
- Submit the following:
 - Screenshot of the original error
 - Reason and justification for the error
 - Corrected code
 - Screenshot of the successful output

```
public class Test {  
    public static void main(String[] args) {  
  
        System.out.println("Welcome to Java")  
  
    }  
}
```

```
public class Test {  
  
    public static void main(String args) {  
  
        System.out.println("Java Programming");  
    }  
}
```

```
public class Student {  
  
    public static void main(String[] args) {  
  
        int student-name = 101;  
  
        System.out.println(student-name);  
    }  
}
```

```
public class Test {  
  
    public static void main(String[] args) {  
  
        int price = 99.50;  
  
        System.out.println(price);  
    }  
}
```

```
public class Test {  
  
    public static void main(String[] args) {  
  
        char grade = "A";  
  
        System.out.println(grade);  
    }  
}
```

```
public class Test {  
  
    public static void main(String[] args) {  
  
        String city = 'Anand';  
  
        System.out.println(city);  
    }  
}
```

```
public class Test {  
  
    public static void main(String[] args) {  
  
        int marks;  
  
        System.out.println(marks);  
    }  
}
```

```
public class Test {  
  
    public static void main(String[] args) {  
  
        double price = 500.75;  
  
        int amount = price;  
  
        System.out.println(amount);  
    }  
}
```

```
public class Test {  
  
    public static void main(String[] args) {  
  
        if (true) {  
  
            int marks = 80;  
        }  
  
        System.out.println(marks);  
    }  
}
```

```
public class Test {  
  
    public static void main(String[] args) {  
  
        final double GST = 18.0;  
  
        GST = 20.0;  
  
        System.out.println(GST);  
    }  
}
```

```
public class Test {  
  
    public static void main(String[] args) {  
  
        int a = 10;  
        int b = 20;  
        int c = 30;  
  
        double average = a + b + c / 3;  
  
        System.out.println("Average = " + average);  
    }  
}
```

```
public class Test {  
  
    public static void main(String[] args) {  
  
        int total = 5;  
        int students = 2;  
  
        double average = total / students;  
  
        System.out.println(average);  
    }  
}
```

```
public class Test {  
  
    public static void main(String[] args) {  
  
        int age = 20;  
  
        if (age = 18) {  
            System.out.println("Eligible");  
        }  
    }  
}
```

```
public class Test {  
  
    public static void main(String[] args) {  
  
        int marks = 75;  
  
        if (marks >= 40 || marks <= 100) {  
            System.out.println("Valid Marks");  
        }  
    }  
}
```

Task: The condition should accept only marks between 40 and 100.

```
public class Test {  
  
    public static void main(String[] args) {  
  
        int count = 5;  
  
        count++;  
  
        System.out.println(count);  
    }  
}
```

```
public class Result {  
  
    public static void main(String[] args) {  
  
        int marks = 75;  
  
        if (marks >= 90)  
            System.out.println("A+");  
        else if (marks >= 60)  
            System.out.println("B");  
        else  
            System.out.println("C");  
    }  
}
```

```
public class Test {  
  
    public static void main(String[] args) {  
  
        int marks = 80;  
  
        if (marks >= 40)  
            System.out.println("Pass");  
            System.out.println("Eligible");  
        else  
            System.out.println("Fail");  
    }  
}
```

```
public class Day {  
  
    public static void main(String[] args) {  
  
        int day = 2;  
  
        switch (day) {  
  
            case 1:  
                System.out.println("Monday")  
  
            case 2:  
                System.out.println("Tuesday");  
                break;  
  
            default:  
                System.out.println("Invalid");  
  
        }  
    }  
}
```

```
public class Test {  
  
    public static void main(String[] args) {  
  
        int i = 1;  
  
        while (i <= 5) {  
  
            System.out.println(i);  
  
        }  
    }  
}
```

```
public class Test {  
  
    public static void main(String[] args) {  
  
        for (int i = 1; i <= 5; i--) {  
  
            System.out.println(i);  
  
        }  
    }  
}
```

Task: Print numbers from 1 to 5.

```
public class Test {  
  
    public static void main(String[] args) {  
  
        int sum;  
  
        for (int i = 1; i <= 10; i++) {  
  
            sum = sum + i;  
        }  
  
        System.out.println(sum);  
    }  
}
```

```
public class Test {  
  
    public static void main(String[] args) {  
  
        for (int i = 1; i <= 10; i++) {  
  
            if (i == 5);  
                break;  
  
            System.out.println(i);  
        }  
    }  
}
```

```
public class Factorial {  
  
    public static void main(String[] args) {  
  
        int n = 5;  
        int fact = 0;  
  
        for (int i = 1; i <= n; i++) {  
  
            fact = fact * i;  
        }  
  
        System.out.println("Factorial = " + fact);  
    }  
}
```

```
public class Test {  
  
    public static void main(String[] args) {  
  
        int marks[5];  
  
        marks[0] = 80;  
  
        System.out.println(marks[0]);  
    }  
}
```

```
public class Test {  
  
    public static void main(String[] args) {  
  
        int[] marks = {70, 80, 90, 85, 75};  
  
        System.out.println(marks[5]);  
    }  
}
```

```
public class Test {  
  
    public static void main(String[] args) {  
  
        int[] numbers = {10, 20, 30, 40};  
  
        int sum = 0;  
  
        for (int i = 0; i < numbers.length; i++) {  
  
            sum = numbers[i];  
        }  
  
        System.out.println("Sum = " + sum);  
    }  
}
```

Task: Correct the program so that it calculates the total.

```
public class Test {  
  
    public static void main(String[] args) {  
  
        int[] marks = {65, 82, 91, 74, 88};  
  
        int max = 0;  
  
        for (int i = 0; i < marks.length; i++) {  
  
            if (marks[i] > max) {  
                max = marks[i];  
            }  
        }  
  
        System.out.println("Highest = " + max);  
    }  
}
```

```
class Student {  
  
    String name;  
  
    void display() {  
  
        System.out.println(name);  
    }  
}  
  
public class Test {  
  
    public static void main(String[] args) {  
  
        Student.display();  
    }  
}
```

```
class Calculator {  
  
    int add(int a, int b) {  
  
        return a + b;  
    }  
}  
  
public class Test {  
  
    public static void main(String[] args) {  
  
        Calculator c = new Calculator();  
  
        System.out.println(c.add(10));  
    }  
}
```

```
class Calculator {  
  
    int square(int n) {  
  
        System.out.println(n * n);  
    }  
}  
  
public class Test {  
  
    public static void main(String[] args) {  
  
        Calculator c = new Calculator();  
  
        System.out.println(c.square(5));  
    }  
}
```

```
class Student {  
  
    String name;  
  
    Student(String name) {  
  
        name = name;  
    }  
  
    void display() {  
  
        System.out.println(name);  
    }  
}  
  
public class Test {  
  
    public static void main(String[] args) {  
  
        Student s = new Student("Rahul");  
  
        s.display();  
    }  
}
```

Task: Correct the constructor so that the object's `name` is initialized.

```
class Calculator {  
  
    int add(int a, int b) {  
        return a + b;  
    }  
  
    int add(int x, int y) {  
        return x + y;  
    }  
}  
  
public class Test {  
  
    public static void main(String[] args) {  
  
        Calculator c = new Calculator();  
  
        System.out.println(c.add(10, 20));  
    }  
}
```

```
class Student {  
  
    String name;  
  
    void Student(String name) {  
  
        this.name = name;  
    }  
}  
  
public class Test {  
  
    public static void main(String[] args) {  
  
        Student s = new Student("Amit");  
  
        System.out.println(s.name);  
    }  
}
```

```
class Student {  
  
    String name;  
  
    Student(String name) {  
  
        this.name = name;  
    }  
}  
  
public class Test {  
  
    public static void main(String[] args) {  
  
        Student s = new Student();  
  
        System.out.println(s.name);  
    }  
}
```

```
class Student {  
  
    String name;  
  
    static int count = 0;  
  
    Student(String name) {  
  
        this.name = name;  
  
        count++;  
    }  
  
    void display() {  
  
        System.out.println(name);  
    }  
}  
  
public class Test {  
  
    public static void main(String[] args) {  
  
        Student s1 = new Student("A");  
        Student s2 = new Student("B");  
  
        System.out.println(s1.count);  
        System.out.println(s2.count);  
    }  
}
```

Task: Identify whether there is an error. If not, improve the code so that `count` is accessed correctly.

```
class Student {  
  
    String name = "Rahul";  
  
    static void display() {  
  
        System.out.println(name);  
    }  
}  
  
public class Test {  
  
    public static void main(String[] args) {  
  
        Student.display();  
    }  
}
```

```
class Product {  
  
    final double GST = 18;  
  
    void changeGST() {  
  
        GST = 20;  
    }  
}  
  
public class Test {  
  
    public static void main(String[] args) {  
  
        Product p = new Product();  
  
        p.changeGST();  
    }  
}
```

```
class Vehicle {  
  
    void start() {  
  
        System.out.println("Vehicle Started");  
    }  
}  
  
class Bike Vehicle {  
  
    void ride() {  
  
        System.out.println("Bike Riding");  
    }  
}  
  
public class Test {  
  
    public static void main(String[] args) {  
  
        Bike b = new Bike();  
  
        b.start();  
        b.ride();  
    }  
}
```

```
class Person {  
  
    Person(String name) {  
  
        System.out.println(  
            "Name: " + name);  
    }  
}  
  
class Student extends Person {  
  
    Student() {  
  
        System.out.println(  
            "Student Created");  
    }  
}  
  
public class Test {  
  
    public static void main(String[] args) {  
  
        Student s = new Student();  
    }  
}
```

```
class Person {  
  
    String name = "Raj";  
}  
  
class Student extends Person {  
  
    String name = "Amit";  
  
    void display() {  
  
        System.out.println(  
            super.name());  
    }  
}  
  
public class Test {  
  
    public static void main(String[] args) {  
  
        Student s = new Student();  
  
        s.display();  
    }  
}
```

```
class Vehicle {  
  
    void start() {  
  
        System.out.println(  
            "Vehicle starts");  
    }  
}  
  
class Bike extends Vehicle {  
  
    void start() {  
  
        System.out.println(  
            "Bike starts");  
  
        super.start();  
    }  
}  
  
public class Test {  
  
    public static void main(String[] args) {  
  
        Bike b = new Bike();  
  
        b.start();  
    }  
}
```

Task: Determine whether there is an error. If yes, correct it; otherwise explain why it is valid.

```
class Animal {  
  
    void sound() {  
  
        System.out.println(  
            "Animal Sound");  
    }  
}  
  
class Dog extends Animal {  
  
    int sound() {  
  
        System.out.println(  
            "Dog Bark");  
  
        return 1;  
    }  
}  
  
public class Test {  
  
    public static void main(String[] args) {  
  
        Dog d = new Dog();  
  
        d.sound();  
    }  
}
```

```
class Animal {  
  
    void sound() {  
  
        System.out.println(  
            "Animal Sound");  
    }  
}  
  
class Dog extends Animal {  
  
    void sound() {  
  
        System.out.println(  
            "Dog Bark");  
    }  
}  
  
public class Test {  
  
    public static void main(String[] args) {  
  
        Animal a = new Animal();  
  
        a = new Dog();  
  
        a.sound();  
    }  
}
```

Task: Determine whether this program contains an error. If not, explain why `Dog Bark` is printed.

```
abstract class Vehicle {  
  
    abstract void start();  
}  
  
class Bike extends Vehicle {  
  
    void start() {  
  
        System.out.println(  
            "Bike Started");  
    }  
}  
  
public class Test {  
  
    public static void main(String[] args) {  
  
        Vehicle v = new Vehicle();  
  
        v.start();  
    }  
}
```

Task: Correct the object creation.

```
abstract class Animal {  
  
    abstract void sound();  
}  
  
class Dog extends Animal {  
  
}  
  
public class Test {  
  
    public static void main(String[] args) {  
  
        Dog d = new Dog();  
  
        d.sound();  
    }  
}
```

```
interface Payment {  
  
    void pay(double amount);  
}  
  
class UPI implements Payment {  
  
    void pay(double amount) {  
  
        System.out.println(  
            "Paid Rs. " + amount + " using UPI");  
    }  
}  
  
public class Test {  
  
    public static void main(String[] args) {  
  
        Payment p = new UPI();  
  
        p.pay(500);  
    }  
}
```